TASK 1 - REPORT

California Housing Price Prediction

1. Objective

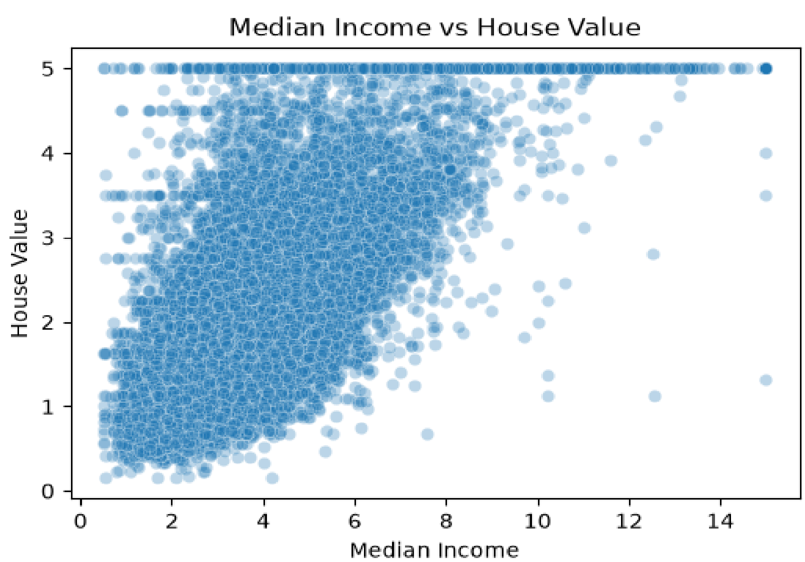
This project builds and evaluates a linear regression model to predict median house prices across California districts, using the California Housing dataset built into scikit-learn.

2. Dataset Overview

The dataset contains 20,640 records with 8 features: median income, house age, average rooms, average bedrooms, population, average occupancy, latitude, and longitude. The target variable is median house value. No missing values were found in any column.

3. Exploratory Data Analysis

Correlation analysis showed median income has the strongest relationship with house price (correlation = 0.69), followed by average rooms (0.15) and house age (0.11). Latitude showed a weak negative correlation (-0.14), suggesting prices tend to decrease slightly moving north in California. A scatter plot of income vs. house value confirmed this positive relationship visually.



4. Methodology

The dataset was split into 80% training data (16,512 rows) and 20% testing data (4,128 rows). A Linear Regression model was trained on the training set and evaluated on the unseen test set to measure how well it generalizes to new data.

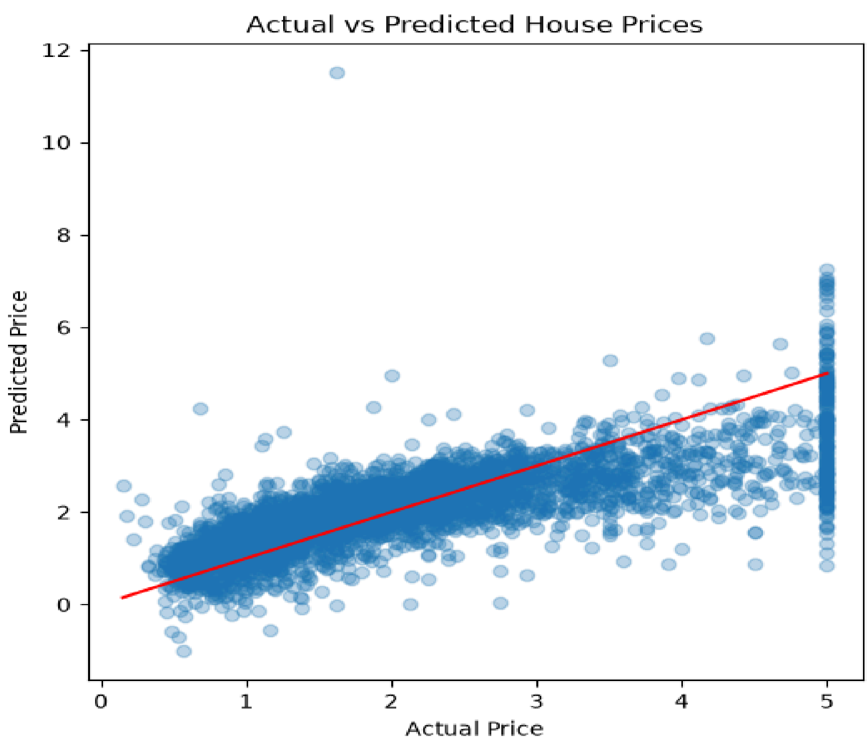
5. Results

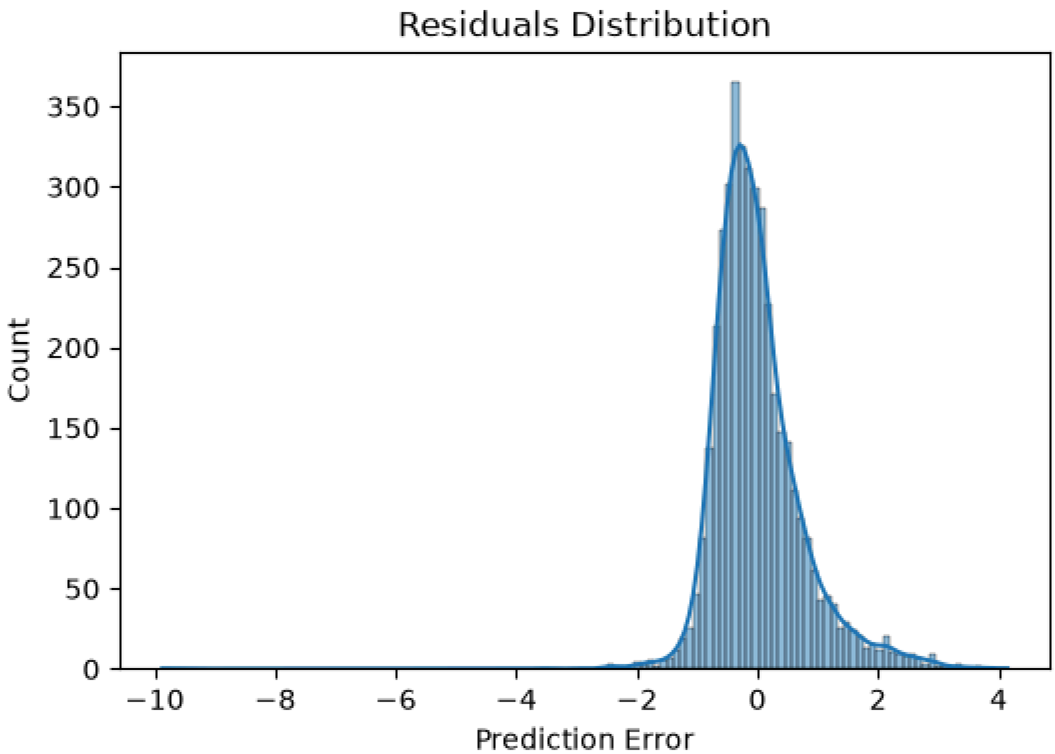
- R^2 Score: 0.576 (the model explains approximately 58% of the variation in house prices)

- MAE: 0.533 (approx. $53,300 average prediction error)

- RMSE: 0.746 (approx. $74,600, weighted more heavily toward larger errors)

An Actual vs. Predicted plot showed reasonably strong alignment for mid-range prices, with more scatter at higher price points. A residual distribution plot showed errors clustered close to zero, with a slight right skew.





6. Limitations & Improvement Ideas

- House prices in this dataset are capped at $500,000, meaning any home actually worth more is recorded as exactly $500,000. This creates a visible cluster of capped values and likely limits model accuracy for high-value properties.

- One prediction fell far outside the realistic price range, suggesting the model is sensitive to unusual combinations of feature values.Outlier detection before training could improve robustness.

- Linear regression assumes straight-line relationships between features and price. A tree-based model such as Random Forest could likely capture more complex, non-linear patterns and improve the R^2 score.

7. Conclusion

This project demonstrates a complete machine learning workflow — data loading, exploratory analysis, model training, evaluation, and visualization — using linear regression as a baseline model. While the model achieves reasonable accuracy, the identified limitations point to clear next steps for improving performance in future iterations.